

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 7: Ratio and Proportion

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Properties of proportion

1. Use the properties of proportion to solve $\frac{6x^2 + 3x - 5}{3x - 5} = \frac{9x^2 + 2x + 5}{2x + 5}$, where $x \neq 0$.

ICSE 2025, 3 marks

SOLUTION

Apply dividendo: if $\frac{a}{b} = \frac{c}{d}$ then $\frac{a-b}{b} = \frac{c-d}{d}$.

$$\frac{6x^2 + 3x - 5 - (3x - 5)}{3x - 5} = \frac{9x^2 + 2x + 5 - (2x + 5)}{2x + 5}, \text{ so } \frac{6x^2}{3x - 5} = \frac{9x^2}{2x + 5}$$

Since $x \neq 0$, divide both sides by $3x^2$: $\frac{2}{3x - 5} = \frac{3}{2x + 5}$

$$4x + 10 = 9x - 15, \text{ so } 5x = 25.$$

Answer: $x = 5$

2. Given $\frac{(a+b)^3}{(a-b)^3} = \frac{64}{27}$, find (a) $\frac{a+b}{a-b}$ (b) the ratio $a : b$, using the properties of proportion.

ICSE 2024, 4 marks

SOLUTION

(a) Take the cube root of both sides: $\frac{a+b}{a-b} = \frac{4}{3}$

(b) Apply componendo and dividendo: $\frac{(a+b) + (a-b)}{(a+b) - (a-b)} = \frac{4+3}{4-3}$

$$\frac{2a}{2b} = \frac{7}{1}$$

Answer: (a) $\frac{4}{3}$ (b) $a : b = 7 : 1$

3. Use componendo and dividendo to solve $\frac{\sqrt{2x+2} + \sqrt{2x-1}}{\sqrt{2x+2} - \sqrt{2x-1}} = 3$. ICSE 2023, 3 marks

SOLUTION

Write 3 as $\frac{3}{1}$ and apply componendo and dividendo: $\frac{2\sqrt{2x+2}}{2\sqrt{2x-1}} = \frac{3+1}{3-1} = 2$

Square both sides: $\frac{2x+2}{2x-1} = 4$, so $2x+2 = 8x-4$

$$6x = 6$$

Answer: $x = 1$

4. Find $x : y$, given $\frac{x^2 + 2x}{2x + 4} = \frac{y^2 + 3y}{3y + 9}$. **ICSE 2020, 3 marks**

SOLUTION

Factorise: $\frac{x(x + 2)}{2(x + 2)} = \frac{y(y + 3)}{3(y + 3)}$

$\frac{x}{2} = \frac{y}{3}$. By alternendo, $\frac{x}{y} = \frac{2}{3}$.

Answer: $x : y = 2 : 3$

5. If $x = \frac{\sqrt{2a+1} + \sqrt{2a-1}}{\sqrt{2a+1} - \sqrt{2a-1}}$, prove that $x^2 - 4ax + 1 = 0$. **ICSE 2020, 4 marks**

SOLUTION

Write x as $\frac{x}{1}$ and apply componendo and dividendo: $\frac{x+1}{x-1} = \frac{2\sqrt{2a+1}}{2\sqrt{2a-1}}$

Square both sides: $\frac{x^2 + 2x + 1}{x^2 - 2x + 1} = \frac{2a+1}{2a-1}$

Apply componendo and dividendo again: $\frac{2x^2 + 2}{4x} = \frac{4a}{2}$, so $\frac{x^2 + 1}{2x} = 2a$

$x^2 + 1 = 4ax$, that is $x^2 - 4ax + 1 = 0$. **Proved.**

2 Proportion

6. A train moves at a uniform speed. It covers 60 m in 2 s, 90 m in x s and y m in 5 s. Find x and y . (a) $x = 4$, $y = 150$ (b) $x = 3$, $y = 100$ (c) $x = 4$, $y = 100$ (d) $x = 3$, $y = 150$ **ICSE 2024, 1 mark**

SOLUTION

Speed = $\frac{60}{2} = 30$ m/s. At uniform speed, distance is proportional to time.

$x = \frac{90}{30} = 3$ and $y = 30 \times 5 = 150$

Answer: (d)

7. What number must be added to each of 4, 6, 8 and 11 so that the four numbers are in proportion? **ICSE 2023, 3 marks**

SOLUTION

Let the number be x . Then $\frac{4+x}{6+x} = \frac{8+x}{11+x}$.

$(4+x)(11+x) = (6+x)(8+x)$, so $44 + 15x + x^2 = 48 + 14x + x^2$

$x = 4$. Check: 8, 10, 12, 15 gives $8 \times 15 = 10 \times 12 = 120$.

Answer: 4

3 Continued proportion

8. Find the mean proportional between 4 and 9. (a) 4 (b) 6 (c) 9 (d) 36 **ICSE 2023, 1 mark**

SOLUTION

$$\text{Mean proportional} = \sqrt{4 \times 9} = \sqrt{36} = 6$$

Answer: (b)

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